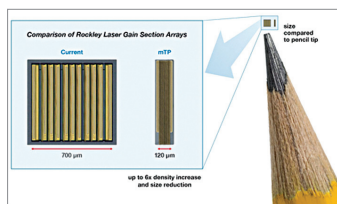




additional reductions in chip manufacturing costs. Silicon-photonics based bio-sensing chips can help designers to make compact devices for the detection



and measurement of multiple biomarkers. These can also be used for creating ultra-small wearables, clothing, or XR/VR/AR headsets and glasses.

Rockley Photonics

<https://rockleyphotonics.com>

Motor driver IC with built-in constant-current control

TB67S549FTG from Toshiba is a stepping motor driver IC with a built-in constant-current control. It can enable motor control without requiring any external circuit components. The motor driver IC is rated for 40V, up to 1.5A, and comes in a QFN24 package, which saves up to 64% space than previous-generation ICs. The IC is suitable for automotive application and supports motor power supply voltages from 4.5V to 33V, and has a current consumption of up to 1µA in sleep mode, making it widely usable for 12V/24V power supply applications. The small size of IC and no requirement for external components reduces power consumption, bill of material, and cost, thus simplifying the designing process.

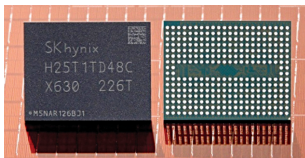
Toshiba

<https://toshiba.semicon-storage.com>

World's highest 238-layer TLC 4D NAND flash

SK hynix has announced the world's highest 238-layer 512Gb TLC 4D NAND

flash. The triple level cell (TLC) NAND offers high productivity, efficiency, smaller size, and faster data transfer speed. It offers 34% higher density per unit area compared to a 176-layer NAND and features a data transfer speed of 2.4Gb per second, which is around 50% higher than the previous-



generation NAND. The smaller size and high efficiency of this flash can help design compact devices and enable faster speed and higher data transfer rate in upcoming smartphones, tablets, and notebook PCs.

SK hynix

<https://news.skhynix.com>

Tiny standalone GNSS module with high tracking sensitivity

ST Microelectronics' Teseo-LIV3FL is a tiny standalone global navigation satellite system (GNSS) module that works with a standard 3.3V input; it can also



work with 1.8V input. The module employs an on-board temperature compensated

crystal oscillator, which enables a very high tracking sensitivity of -163dBm, providing 1.5-metre CEP position accuracy. The GNSS module offers simultaneous multi-constellation tracking, including GPS, Galileo, Glonass, BeiDou, and QZSS. It includes seven-day autonomous assisted GNSS, predictive assisted GNSS, and real-time assisted GNSS. The module comes in a small package measuring only 9.7mm × 10.1mm, making it suitable for use in compact products and in applications where discrete tracking is vital, such as goods tracking, drones, tolling, anti-theft systems, people and pet location.

ST Microelectronics

<https://www.st.com>

Industry's fastest DRAM ever

Samsung has announced the world's fastest LPDDR5X DRAM that offers a speed of 8.5Gbps, which is more than 10% faster than the current fastest DRAM available in the market. It offers high performance, lower power,



and compact size. The high-speed memory can enable newer applications in various fields, such as AI and the metaverse. It can also improve the performance of smartphones, tablets, PCs, etc. The DRAM's low-power characteristics can help reduce power consumption levels, and its ability to process massive amounts of vehicle data at much faster speeds makes it suitable for applications in autonomous vehicles.

Samsung

<https://news.samsung.com>

World's first RISC-V CPU with ISO 26262 compliance

The N25F-SE is a 32-bit RISC-V CPU core that supports standard IMACFD extensions, including efficient integer instructions and single/double precision floating point instructions.

The processor from Andes is the world's first RISC-V CPU with ISO 26262 compliance. It employs an efficient 5-stage pipeline, which provides a good balance of high operating frequency and compact design. The development process and safety design of N25F-SE can reduce systematic failures and random hardware failures. The CPU comes with a ASIL B certification and can be incorporated in both new electronic systems or existing systems. It can be used in a wide range of applications, including in-car monitoring, keyless entry, lighting control, tire pressure monitoring, vision ADAS, etc.

Andes Technology

<http://www.andestech.com>